Machine Learning Curriculum and Roadmaps

Unit 1: Linear Algebra and Coordinate Spaces

Linear Algebra serves as the absolute mathematical foundation for representing data in machine learning. Datasets are represented as high-dimensional matrices, where each row corresponds to a data point and each column represents a specific feature dimension.

Key concepts include Vectors, which define positions in multi-dimensional space, and Vector Spaces, which govern how these vectors scale and combine. Linear Transformations allow us to map, rotate, and project data from one coordinate space to another. Specifically, Eigenvalues and Eigenvectors are critical for finding directions of maximum variance in dataset mapping algorithms like Principal Component Analysis (PCA). Understanding vectors and matrices is the prerequisite for calculating multidimensional equations.

Unit 2: Calculus and Optimization Gradients

Calculus provides the mathematical machinery to train machine learning models by optimizing their weights. While Linear Algebra represents the parameters of a model, Calculus allows us to adjust those parameters to reduce prediction errors.

The core concept is the Derivative, which measures the rate of change of a function. In multidimensional machine learning, we calculate Partial Derivatives to find changes relative to individual model weights. By compiling these partial derivatives, we obtain the Gradient vector. The Gradient points in the direction of steepest increase of the error function. Consequently, algorithms use Gradient Descent to step in the opposite direction of the gradient to minimize prediction error. You must understand Derivatives and Linear Algebra before studying Gradient Descent and optimization.

Unit 3: Probability Theory and Bayes Theorem

Probability Theory is the mathematical framework for modeling uncertainty and making predictions in machine learning. Real-world data is noisy and incomplete, requiring probabilistic models to quantify confidence.

We use Probability Distributions (like Gaussian or Binomial distributions) to describe how likely different data outcomes are. Bayes Theorem is the fundamental law of conditional probability, allowing us to update the probability of a hypothesis as we observe new evidence. Bayes Theorem is the mathematical prerequisite for probability classification models like Naive Bayes classifiers. Together, Linear Algebra, Calculus, and Probability Theory form the complete mathematics prerequisite to learn Machine Learning.

Unit 4: Machine Learning Foundations

Machine Learning is the field of computer science that builds algorithms capable of learning from data. It leverages Linear Algebra, Calculus, and Probability Theory to detect patterns and make predictions.

Supervised Learning is the training paradigm where models learn from labeled datasets, mapping inputs to target outputs (for example, in Regression models predicting housing prices or Classification models predicting image labels). Unsupervised Learning discovers hidden patterns and clusters in unlabeled data. Understanding basic Machine Learning classification concepts is the prerequisite to advance to Neural Networks.

Unit 5: Neural Networks and Backpropagation

Neural Networks are a class of machine learning models inspired by the biological brain, forming the foundation of deep learning. A Neural Network consists of layers of nodes (neurons): an input layer, one or more hidden layers, and an output layer.

During the forward pass, data flows through weights and activation functions to make a prediction. The core optimization technique is Backpropagation, which uses the calculus chain rule to compute the error gradient with respect to all model weights. These weights are then updated using Gradient Descent. Prerequisites for Neural Networks include general Machine Learning concepts, Calculus derivatives, and Linear Algebra matrix operations. Mastery of basic Neural Networks is required before moving to specialized architectures like CNNs and RNNs.

Unit 6: Convolutional and Recurrent Neural Networks

Once you understand basic Neural Networks, you can study specialized deep architectures optimized for different data formats.

Convolutional Neural Networks (CNNs) are specialized networks designed to process grid-like spatial data, such as digital images, using kernel filters to extract features. Recurrent Neural Networks (RNNs) are specialized for processing sequential data, such as time series or text, by maintaining a temporal loop state. Understanding sequential RNN structures is a key prerequisite to learning the Attention Mechanism and Transformers.

Unit 7: Transformers and Attention Mechanisms

Transformers are the modern state-of-the-art deep learning architectures that have revolutionized how models process sequential data, replacing older sequential RNN structures.

The core mechanism of a Transformer is Self-Attention, which calculates the mathematical alignment between all words in a sentence simultaneously. This allows parallel processing on modern GPUs, overcoming the temporal sequence limits of RNNs. The prerequisite to learn Transformers is a solid understanding of Neural Networks, matrix math from Linear Algebra, and sequential text modeling. Transformers are the primary building block for Large Language Models.

Unit 8: Large Language Models and NLP

Large Language Models (LLMs) represent the cutting edge of Natural Language Processing (NLP). LLMs are massive neural networks trained on vast textual datasets to generate human-like text.

These models are built directly using the Transformer architecture and trained via self-supervised objectives, predicting masked words in a sentence. Key techniques include fine-tuning and prompt engineering. The prerequisite path to understand LLMs goes through Linear Algebra, Calculus, Probability, Machine Learning, Neural Networks, and Transformers.